

EmbedUR Linux Training

Module - 2 : The Linux Environment

Assessment - 2

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Question – 1: Use the appropriate command to list all files larger than 1 MB in the current directory and save the output to a file

```
Activities Terminal Jan 31 10:33 kishore@kishore-virtual-machine: ~/Desktop/Linux Training
kishore@kishore-virtual-machine:~/Desktop$ cd Linux Training
bash: cd: too many arguments
kishore@kishore-virtual-machine:~/Desktop$ cd 'Linux Training'
kishore@kishore-virtual-machine:~/Desktop/Linux Training$
kishore@kishore-virtual-machine:~/Desktop/Linux Training$ ls
2026_Linux_Training_Program.pptx 'Linux Modules - PPT'
'Campus Hires 2026 - Linux - Assessment Module 1-5.pptx' 'Linux Technical Session'
kishore@kishore-virtual-machine:~/Desktop/Linux Training$ cd 'Linux Modules - PPT'
kishore@kishore-virtual-machine:~/Desktop/Linux Training/Linux Modules - PPT$
kishore@kishore-virtual-machine:~/Desktop/Linux Training/Linux Modules - PPT$ ls
2025_Linux_Training_Program_Module_1.pptx Linux_Training_Program_Module_4.pptx Module6_DriverFundamentals.pptx
2025_Linux_Training_Program_Module_2.pptx Module_5_Linux_Training.pptx
Linux_Training_Program_Module_3.pptx Module6_2_LinuxNSS.pptx
kishore@kishore-virtual-machine:~/Desktop/Linux Training/Linux Modules - PPT$ find . -type f -size +1M
kishore@kishore-virtual-machine:~/Desktop/Linux Training/Linux Modules - PPT$ cd ../
kishore@kishore-virtual-machine:~/Desktop/Linux Training$
kishore@kishore-virtual-machine:~/Desktop/Linux Training$ find . -type f -size +1M
./Linux Technical Session/Campus Hires 2025 Batch - Linux Training Program - Technical Session 2-20250123_105859-Meeting Recording.mp4
./Linux Technical Session/Campus Hires 2025 Batch - Linux Training Program - Technical Session 4-20250129_110417-Meeting Recording.mp4
./Linux Technical Session/Campus Hires 2025 Batch - Linux Training Program - Technical Session 6 (part 1)-20250205_110441-Meeting Recording.mp4
./Linux Technical Session/Campus Hires 2025 Batch - Linux Training Program - Technical Session 3-20250127_110037-Meeting Recording.mp4
./Linux Technical Session/Campus Hires 2025 Batch - Linux Training Program - Technical Session 5-20250131_110141-Meeting Recording.mp4
./Linux Technical Session/Campus Hires 2025 Batch - Linux Training Program - Technical Session 6 (part 2)-20250210_110404-Meeting Recording.mp4
./Linux Technical Session/Campus Hires 2025 Batch - Linux Training Program - Technical Session 1-20250120_110613-Meeting Recording.mp4
kishore@kishore-virtual-machine:~/Desktop/Linux Training$
```

Question – 2: Replace all occurrences of "localhost" with "127.0.0.1" in a configuration file named config.txt, and save the updated file as updated_config.txt.

```
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Training/Module - 2/Question - 2
sed: -e expression #1, char 21: unterminated 's' command
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2/Question - 2$ sed 's/localhost/127.0.0.1/g' config.txt > updated_config.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2/Question - 2$ cat updated_config.txt
# Application Configuration File
# Environment: Development

server.name=dev-server
server.host=127.0.0.1
server.port=8080

database.type=mysql
database.host=127.0.0.1
database.port=3306
database.user=root
database.password=root123

# Backup database server
database.backup.host=127.0.0.1
database.backup.port=3307

# Cache configuration
cache.enabled=true
cache.host=127.0.0.1
cache.port=6379

# Message broker settings
broker.type=rabbitmq
broker.host=127.0.0.1
broker.port=5672
```

Question – 3 : Use the appropriate command to search for lines containing the word "ERROR" in a log file but exclude lines containing "DEBUG". Save the results to a file named filtered_log.txt.

log:

DEBUG: Starting the application initialization.
ERROR: Unable to connect to the database.
INFO: User 'admin' logged in successfully.
DEBUG: Fetching configuration settings from the server.
ERROR: DEBUG - Configuration settings could not be applied.
INFO: Scheduled job 'backup' completed successfully.
DEBUG: Connection to server timed out. Retrying...
ERROR: Failed to fetch data from API endpoint '/users'.
INFO: Maintenance mode activated.

ERROR: DEBUG - Query execution failed due to a syntax error.

DEBUG: Reloading application modules.

ERROR: Missing required parameter in the request.

INFO: Shutting down the system gracefully.

DEBUG: Closing unused network connections.

ERROR: DEBUG - Unexpected server response received.

```
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Training/Module - 2
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2$ touch log.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2$ cat log.txt
DEBUG: Starting the application initialization.
ERROR: Unable to connect to the database.
INFO: User 'admin' logged in successfully.
DEBUG: Fetching configuration settings from the server.
ERROR: DEBUG - Configuration settings could not be applied.
INFO: Scheduled job 'backup' completed successfully.
DEBUG: Connection to server timed out. Retrying...
ERROR: Failed to fetch data from API endpoint '/users'.
INFO: Maintenance mode activated.
ERROR: DEBUG - Query execution failed due to a syntax error.
DEBUG: Reloading application modules.
ERROR: Missing required parameter in the request.
INFO: Shutting down the system gracefully.
DEBUG: Closing unused network connections.
ERROR: DEBUG - Unexpected server response received.

kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2$ grep -in "ERROR" log.txt
2:ERROR: Unable to connect to the database.
5:ERROR: DEBUG - Configuration settings could not be applied.
8:ERROR: Failed to fetch data from API endpoint '/users'.
10:ERROR: DEBUG - Query execution failed due to a syntax error.
12:ERROR: Missing required parameter in the request.
15:ERROR: DEBUG - Unexpected server response received.
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2$ grep -in "ERROR" log.txt > filtered_log.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2$ cat filtered_log.txt
2:ERROR: Unable to connect to the database.
5:ERROR: DEBUG - Configuration settings could not be applied.
8:ERROR: Failed to fetch data from API endpoint '/users'.
10:ERROR: DEBUG - Query execution failed due to a syntax error.
12:ERROR: Missing required parameter in the request.
15:ERROR: DEBUG - Unexpected server response received.
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Module - 2$
```

Question – 4: Write a code to identify the process with the highest memory usage and then terminate it.

Start by giving the top command to get a detailed information about the process that are currently running.

```
Activities  Terminal  Jan 31 10:09
kishore@kishore-virtual-machine: ~/Desktop
kishore@kishore-virtual-machine:~/Desktop$ top
```

Then, press shift + M to sort the table output based on memory usage percentage.

```
Activities Terminal Jan 31 10:21 kishore@kishore-virtual-machine: ~/Desktop

top - 10:20:59 up 33 min, 1 user, load average: 0.09, 0.48, 0.56
Tasks: 337 total, 1 running, 336 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.0 us, 0.3 sy, 0.0 ni, 98.7 id, 0.0 wa, 0.0 hi, 0.0 st, 0.0 st
MiB Mem : 3868.4 total, 182.6 free, 2504.7 used, 1181.1 buff/cache
MiB Swap: 3898.0 total, 3577.5 free, 320.5 used, 924.1 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 2228 kishore    20   0 1711132 440904 139060 S   0.3 11.1   0:30.92 soffice.bin
 4309 kishore    20   0 1393.8g 398648 184216 S   0.3 10.1   0:29.28 chrome
 2473 kishore    20   0  48.9g 369236 218764 S   0.3  9.3   1:26.17 chrome
 4330 kishore    20   0 1392.1g 279980 126256 S   0.0  7.1   0:05.69 chrome
 1612 kishore    20   0 4737128 271336 149624 S   0.3  6.8   2:08.46 gnome-shell
 3727 kishore    20   0 1393.8g 259228 140632 S   0.3  6.5   0:22.18 chrome
 2717 kishore    20   0 1392.2g 240088 135688 S   0.0  6.1   0:35.59 chrome
 3834 kishore    20   0 1392.1g 201624 125952 S   0.0  5.1   0:04.90 chrome
 2664 kishore    20   0 1393.7g 176888 127124 S   0.0  4.5   0:00.69 chrome
 3863 kishore    20   0 1392.1g 155588 117832 S   0.0  3.9   0:00.99 chrome
 4402 kishore    20   0 1392.1g 150332 111464 S   0.0  3.8   0:00.92 chrome
 2516 kishore    20   0  48.7g 143112 110728 S   0.0  3.6   0:58.94 chrome
 2522 kishore    20   0  48.4g 142704 105672 S   0.0  3.6   0:35.16 chrome
 3987 kishore    20   0 1393.7g 124980 88672 S   0.0  3.2   0:03.55 chrome
 3248 kishore    20   0 1393.7g 119332 95104 S   0.0  3.0   0:00.61 chrome
 3437 kishore    20   0 1392.1g 118688 94720 S   0.0  3.0   0:00.30 chrome
 3233 kishore    20   0 1392.1g 109212 88864 S   0.0  2.8   0:00.11 chrome
 3876 kishore    20   0 1392.1g 108768 87040 S   0.0  2.7   0:00.07 chrome
 1726 kishore    20   0 906812 96540 60964 S   0.0  2.4   0:08.95 nautilus
 3704 kishore    20   0  48.6g 82688 67328 S   0.0  2.1   0:00.13 chrome
 4416 kishore    20   0 1392.1g 73572 54528 S   0.0  1.9   0:00.02 chrome
 2490 kishore    20   0  48.4g 64512 50848 S   0.0  1.6   0:00.23 chrome
 2491 kishore    20   0  48.4g 64512 50176 S   0.0  1.6   0:00.21 chrome
 4523 kishore    20   0 1106864 59780 44816 S   0.0  1.5   0:00.36 gnome-calendar
 1654 kishore    20   0 596872 59264 28032 S   0.0  1.5   0:00.89 goa-daemon
 2531 kishore    20   0  48.4g 58444 41356 S   0.0  1.5   0:00.53 chrome
 1790 kishore    20   0 1197812 52848 36592 S   0.0  1.3   0:00.69 evolution-alarm
 2019 kishore    20   0 1141592 51420 33524 S   0.0  1.3   0:02.21 xdg-desktop-por
 2062 kishore    20   0 2876824 49740 35520 S   0.0  1.3   0:01.79 gjs
 2376 kishore    20   0 56852 40804 38004 S   0.3  1.2   0:00.64 gnome-terminal-
 4526 kishore    20   0 393364 41040 25688 S   0.0  1.0   0:00.34 seahorse
 1671 kishore    20   0 1139096 38592 25680 S   0.0  1.0   0:00.64 evolution-calen
 2544 kishore    20   0  48.6g 33100 4736 S   0.0  0.8   0:00.01 chrome
 2001 kishore    20   0 645872 28884 18860 S   0.0  0.7   0:00.11 tracker-miner-f
```

Since the top command is interactive, we can easily customize the output with just a single key. To kill a particular process, the key to be used is K. Press the key K from the keyboard.

```
Activities Terminal Jan 31 10:22 kishore@kishore-virtual-machine: ~/Desktop

top - 10:22:33 up 34 min, 1 user, load average: 0.25, 0.42, 0.53
Tasks: 337 total, 1 running, 336 sleeping, 0 stopped, 0 zombie
%Cpu(s): 5.2 us, 2.0 sy, 0.0 ni, 92.6 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 3868.4 total, 340.5 free, 2437.7 used, 1090.2 buff/cache
MiB Swap: 3898.0 total, 3480.2 free, 417.8 used, 985.0 avail Mem

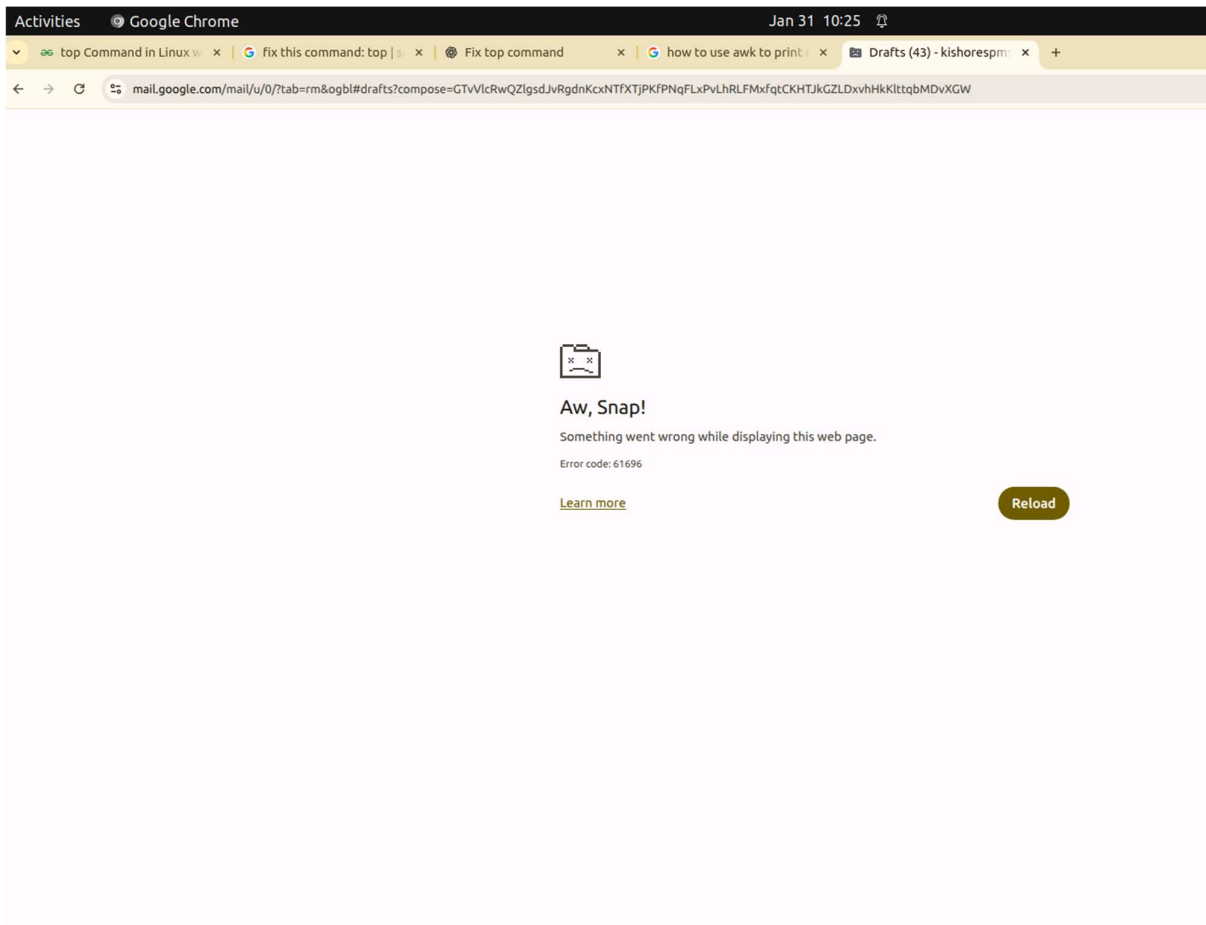
PID to signal/kill [default pid = 2228]
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 2228 kishore    20   0 1721948 425672 139060 S   5.0 10.7   0:48.45 soffice.bin
 4309 kishore    20   0 1393.8g 374128 169468 S   0.7  9.4   0:29.64 chrome
 2473 kishore    20   0  48.8g 366324 216036 S   1.0  9.2   1:26.53 chrome
 4330 kishore    20   0 1392.1g 271404 126256 S   0.3  6.9   0:05.89 chrome
```

Then, there would be a prompt to type in the process id that is required. In our case, type 4309.

```
Activities Terminal Jan 31 10:23 kishore@kishore-virtual-machine: ~/Desktop

top - 10:22:33 up 34 min, 1 user, load average: 0.25, 0.42, 0.53
Tasks: 337 total, 1 running, 336 sleeping, 0 stopped, 0 zombie
%Cpu(s): 5.2 us, 2.0 sy, 0.0 ni, 92.6 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 3868.4 total, 340.5 free, 2437.7 used, 1090.2 buff/cache
MiB Swap: 3898.0 total, 3480.2 free, 417.8 used, 985.0 avail Mem
Send pid 4309 signal [15/sigterm]
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 2228 kishore    20   0 1721948 425672 139060 S   5.0 10.7   0:48.45 soffice.bin
 4309 kishore    20   0 1393.8g 374128 169468 S   0.7  9.4   0:29.64 chrome
 2473 kishore    20   0  48.8g 366324 216036 S   1.0  9.2   1:26.53 chrome
 4330 kishore    20   0 1392.1g 271404 126256 S   0.3  6.9   0:05.89 chrome
```

Waiting for confirmation. Press Enter key



Chrome tab got closed then.

Question – 5: Create 3 files in a dir1 and re-direct the output of list command with sorted by timestamp of the files to a file

```
kishore@kishore-virtual-machine: ~/Desktop
kishore@kishore-virtual-machine:~/Desktop$ netstat -a | sort
Active Internet connections (servers and established)
Active UNIX domain sockets (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
Proto RefCnt Flags   Type       State          I-Node  Path
raw6  0      0  [::]:ipv6-icmp          [::]:*                7
tcp    0      0 localhost:domain        0.0.0.0:*              LISTEN
tcp    0      0 localhost:ipp           0.0.0.0:*              LISTEN
tcp6   0      0 ip6-localhost:ipp      [::]:*                LISTEN
udp    0      0 0.0.0.0:44962          0.0.0.0:*              LISTEN
udp    0      0 0.0.0.0:mdns           0.0.0.0:*              LISTEN
udp    0      0 kishore-virtual-:bootpc 192.168.18.254:bootps  ESTABLISHED
udp    0      0 localhost:domain        0.0.0.0:*              LISTEN
udp6   0      0 [::]:45061             [::]:*                LISTEN
udp6   0      0 [::]:mdns               [::]:*                LISTEN
unix  17      0  [ ]          DGRAM        CONNECTED      10784    /run/systemd/journal/dev-log
unix  2      0  [ ACC ]      SEQPACKET    LISTENING      10790    /run/udev/control
unix  2      0  [ ACC ]      STREAM       LISTENING      10240    /run/systemd/journal/io.systemd.journal
unix  2      0  [ ACC ]      STREAM       LISTENING      10764    /run/systemd/private
unix  2      0  [ ACC ]      STREAM       LISTENING      10766    /run/systemd/userdb/io.systemd.DynamicUser
unix  2      0  [ ACC ]      STREAM       LISTENING      10767    /run/systemd/io.systemd.ManagedOOM
unix  2      0  [ ACC ]      STREAM       LISTENING      10777    /run/systemd/fsck.progress
unix  2      0  [ ACC ]      STREAM       LISTENING      10788    /run/systemd/journal/stdout
unix  2      0  [ ACC ]      STREAM       LISTENING      14843    /run/systemd/resolve/io.systemd.Resolve
unix  2      0  [ ACC ]      STREAM       LISTENING      15438    /var/run/vmware/guestServicePipe
unix  2      0  [ ACC ]      STREAM       LISTENING      15484    /run/acpid.socket
unix  2      0  [ ACC ]      STREAM       LISTENING      15494    /run/avahi-daemon/socket
unix  2      0  [ ACC ]      STREAM       LISTENING      15509    /run/dbus/system_bus_socket
unix  2      0  [ ACC ]      STREAM       LISTENING      15516    /run/snapd.socket
unix  2      0  [ ACC ]      STREAM       LISTENING      15518    /run/snapd-snap.socket
unix  2      0  [ ACC ]      STREAM       LISTENING      15522    /run/uidd/request
unix  2      0  [ ACC ]      STREAM       LISTENING      15659    /run/irabalance/irabalance695.sock
```

```
kishore@kishore-virtual-machine: ~/Desktop
kishore@kishore-virtual-machine:~/Desktop$ sudo tcpdump -d
[sudo] password for kishore:
Warning: assuming Ethernet
(000) ret      #262144
kishore@kishore-virtual-machine:~/Desktop$
```